The Assignment: "The Reality Defender"

(Please note: This test requires only a standard ARCore/ARKit-compatible smartphone, no headset.)

Overview

Time Limit: 2 Days (48 Hours)

Hardware Required: Any AR-compatible Smartphone (Android/iOS)

Engine: Unity (ARFoundation)

Brief: We aren't just looking for code that runs; we are looking for code that *feels good* to play. Your task is to build a simple AR target practice mini-game that demonstrates your grasp of AR stability, visual polish, and gameplay logic.

Part 1: The Arena (AR Interaction)

Your Goal: Demonstrate an understanding of AR Raycasting and Plane Detection.

Your Task:

1. **Setup:** Initialize a Unity project with ARFoundation.
2. **Action:** Detect horizontal planes (floor/table).
3. **Interaction:** Instead of the object automatically appearing, write a script where the user must **Tap the Screen** to spawn the "Game Base" on a detected plane.
4. **Constraint:** The base must align perfectly with the surface normal (it shouldn't look tilted or floating). Once placed, it should be anchored and immovable.

Part 2: The Juice (Shader Graph & VFX)

Your Goal: Show that you understand "Game Feel" and Technical Art. In gaming, feedback is everything.

Your Task:

1. **Do NOT** use standard Unity materials for the targets.
2. **Action:** Create a custom **Shader Graph** for the "Target" objects (spheres/cubes floating above the base).
    - **Requirement:** The targets must look "volatile" or "energy-filled." Use a scrolling noise texture, a fresnel rim light, or a vertex displacement effect (wobbling).
3. **Bonus:** Add a Particle System explosion when a target is destroyed.

Part 3: The Game Loop (C# Scripting)

Your Goal: Implement Physics, Collision, and Game State Logic.

Your Task:

1. **The Weapon:** Turn the User's Main Camera into a cannon. When the user taps a button (create a simple UI button "FIRE"), spawn a projectile from the camera's position moving forward.
2. **The Logic:**
    - **Spawn:** The Game Base (from Part 1) should spawn 3 floating Targets.
    - **Physics:** The projectile must use a Rigidbody to physically collide with the Targets.
    - **State Management:**
        - **Hit:** If projectile hits Target -> Target disappears (with VFX) -> Score +10.
        - **Miss:** If projectile flies > 5 meters -> Destroy projectile (memory management).
3. **UI Feedback:**
    - Display a simple **Score Counter** in world space (floating above the base) or Screen Space (top corner).
    - When all 3 targets are cleared, display "WAVE CLEARED".

Deliverables

1. **Screen Recording:** Show the plane detection, the tap-to-place, and the shooting mechanics (hitting at least 3 targets).
2. **Screenshot:** Your Shader Graph setup.
3. **Please send us all these files as a Google Drive link (with "Anyone with link can View" permission).**